Enhancing Belief Propagation Decoders for Correlated Errors Using Cycle Error Reconstruction

Debankan Sannamoth, **Pavithran Iyer**, and Joseph Emerson

APS March Meeting 2026

MAR-C09:5

16 March 2026

Institute for Quantum Computing

University of Waterloo

email: pavithran.sridhar@gmail.com , web: https://sites.google.com/view/piyer/home

The Big Question …

Quantum Error Correction: Suppress Noise at the logical level with *high overhead cost*

i.i.d noise

Qubits, gates, measurements…

The Big Question …

Quantum Error Correction: Suppress Noise at the logical level with *high overhead cost*

i.i.d noise

Qubits, gates, measurements…

Hardware : errors are often correlated
Revealed using experimental noise characterization (CER technique)

Minimize the overhead using quantum LDPC codes
E.g. ***Hypergraph Product Codes***

The Big Question …

Quantum Error Correction: Suppress Noise at the logical level with *high overhead cost*

i.i.d noise

Qubits, gates, measurements…

Hardware : errors are often correlated
Revealed using experimental noise characterization (CER technique)

Minimize the overhead using quantum LDPC codes
E.g. ***Hypergraph Product Codes***

Tailoring decoding to the underlying noise: Belief Propagation (BP) Decoders for qLDPC

Can we enhance the performance of BP for correlated errors using experimental data?

The Big Question ...

Quantum Error Correction: Suppress Noise at the logical level with *high overhead cost*

i.i.d noise

Qubits, gates, measurements…

Hardware : errors are often correlated
Revealed using experimental noise characterization (CER technique)

Minimize the overhead using quantum LDPC codes
E.g. ***Hypergraph Product Codes***

Tailoring decoding to the underlying noise: Belief Propagation (BP) Decoders for qLDPC

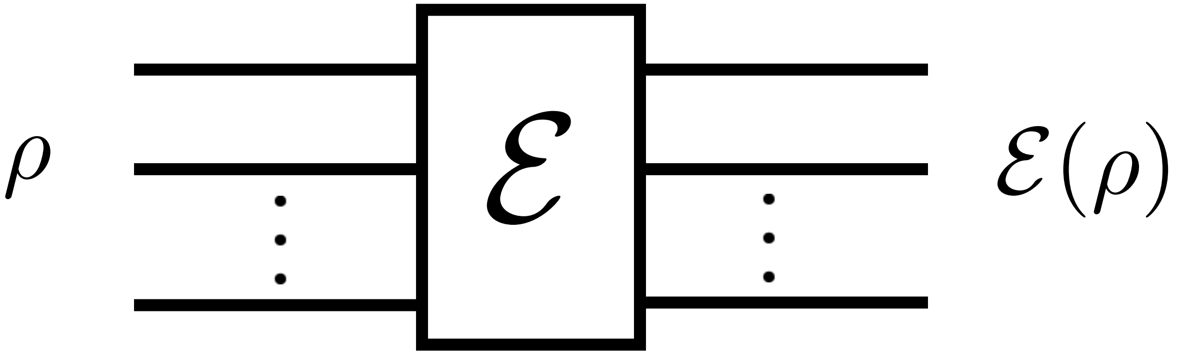
Can we enhance the performance of BP for correlated errors using experimental data?

This talk: **Yes!**

- Constructing a Neural Network from the BP decoder, that adapts to CER data.
- Observe a 10X gain over state-of-the-art BP-OSD decoders for correlated noise.

Cycle Error Reconstruction (CER)

Goal: Given a noisy circuit, extract rates of different Pauli errors.



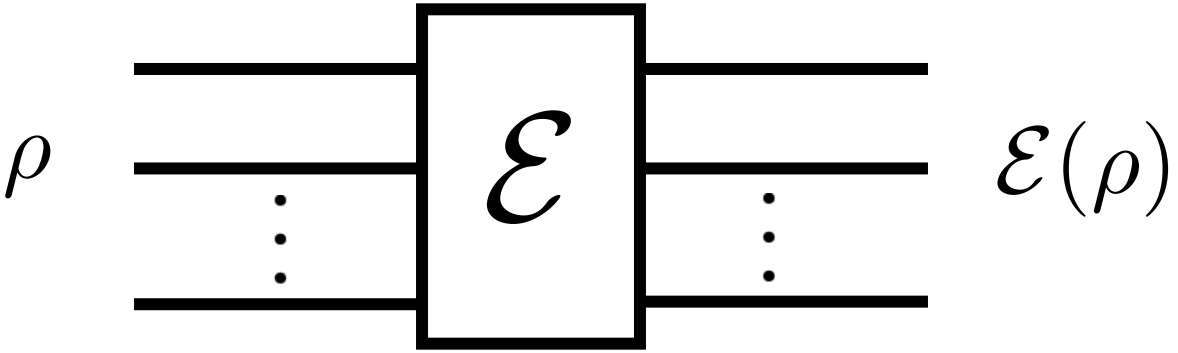
$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \, \rho \, P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

Pauli Errors

Cycle Error Reconstruction (CER)

Goal: Given a noisy circuit, extract rates of different Pauli errors.

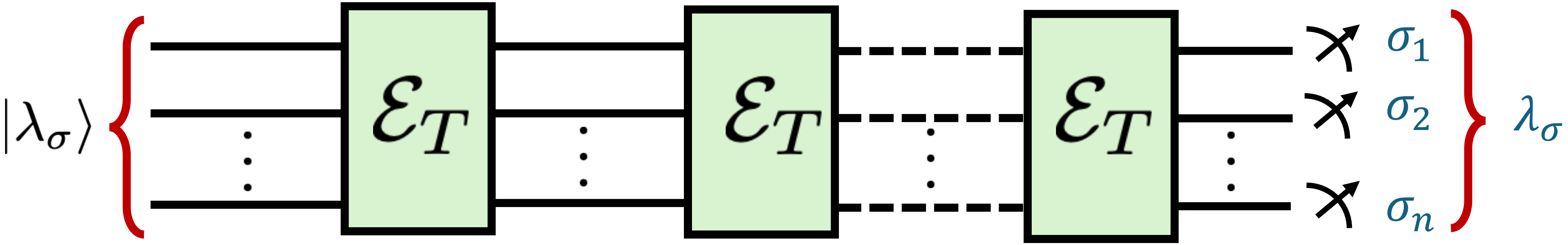


$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \, \rho \, P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

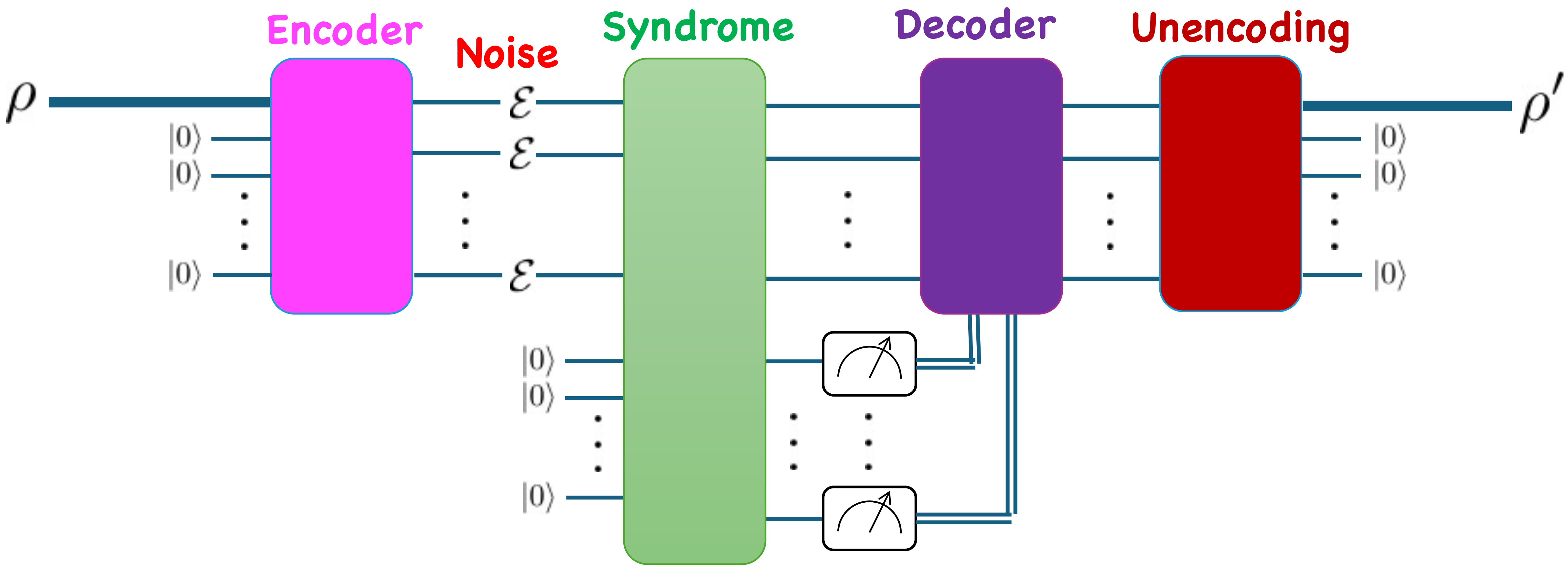
Pauli Errors

Cycle Error Reconstruction: Experimental protocol to estimate $\chi_{i,i}$ for n −qubit CPTP maps $\mathcal{E}$.

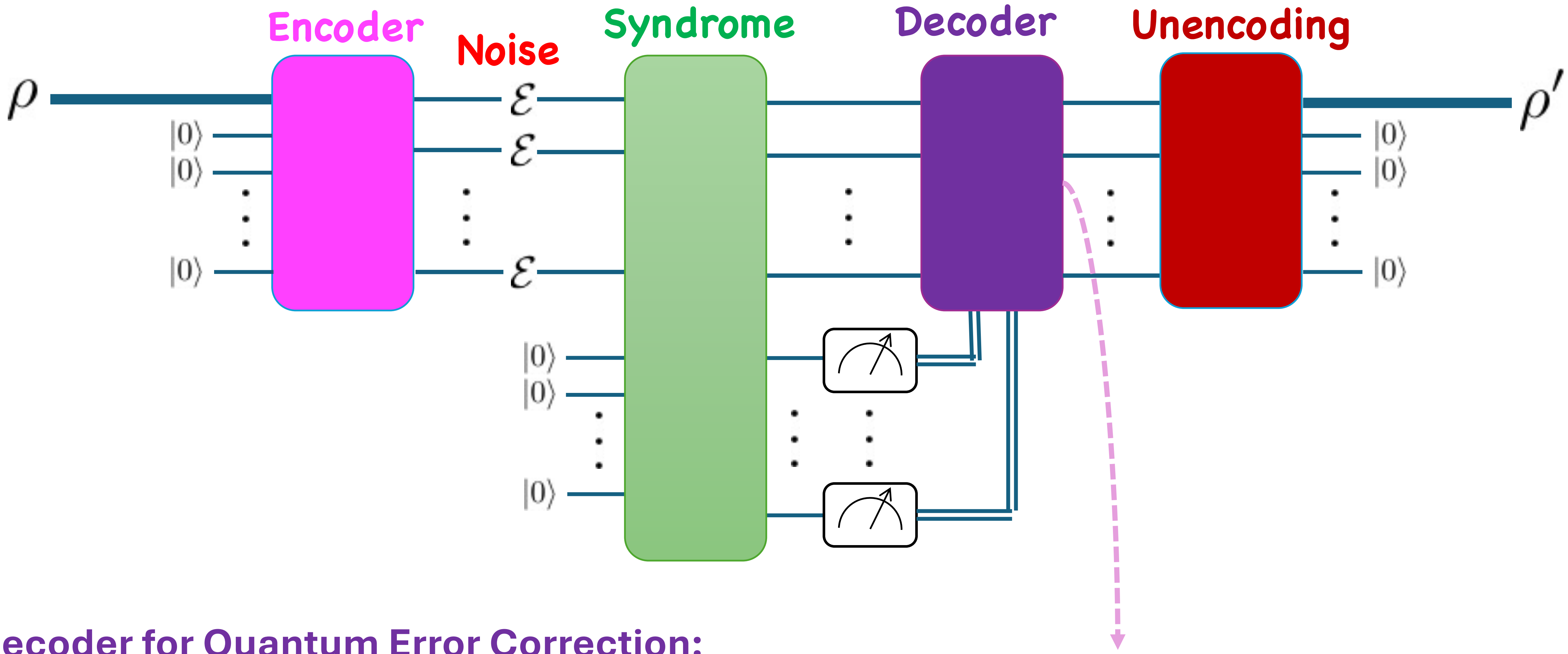


Nat Comm. 10, 5347 (2019), ACM Trans on Quantum Computing, 1, 1 (2020): $\{\lambda_\sigma\} \rightarrow \{\chi_{\sigma,\sigma}\}$

Can CER be useful to a decoder?



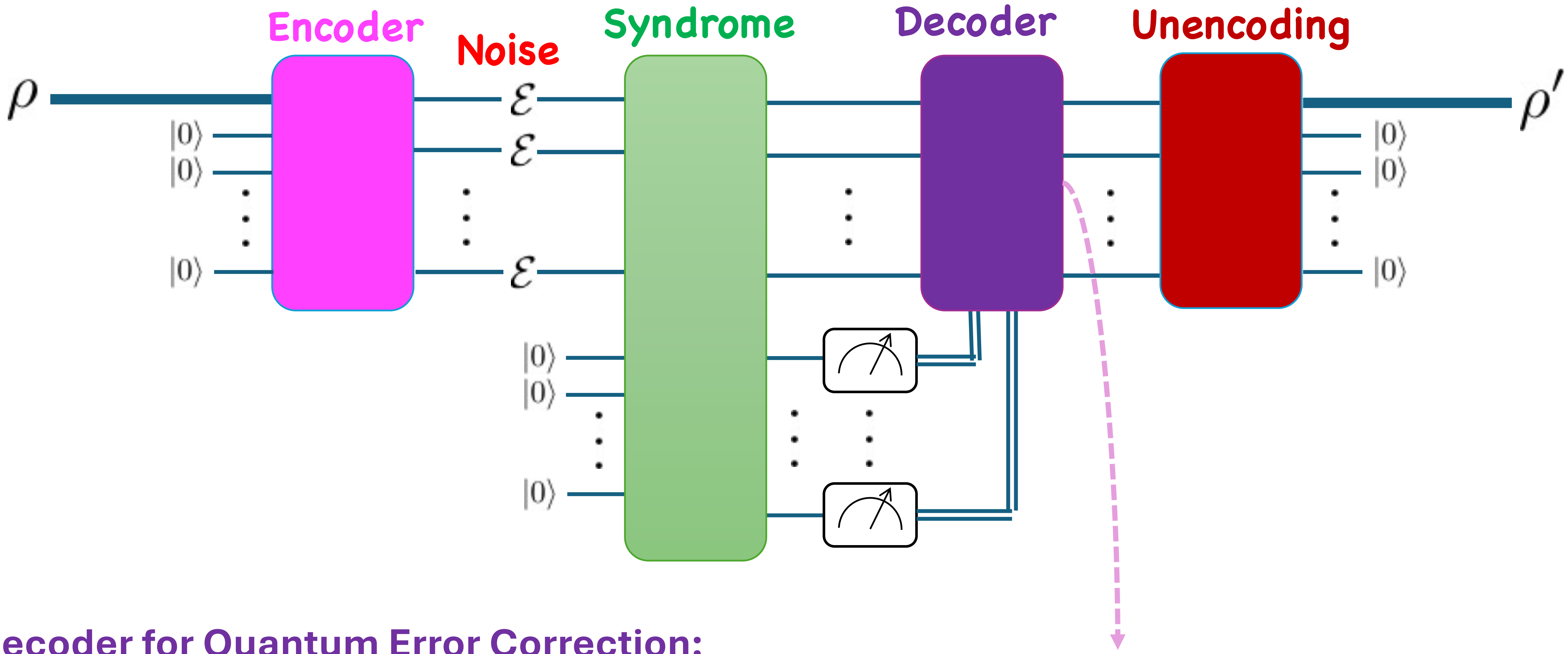
Can CER be useful to a decoder?



Decoder for Quantum Error Correction:

Given an error syndrome s, determine the most like class of Pauli errors that caused it.

Can CER be useful to a decoder?



Decoder for Quantum Error Correction:

Given an error syndrome s, determine the most like class of Pauli errors that caused it.

Can we also provide Pauli decay rates, specifically, single and two-qubit marginals?

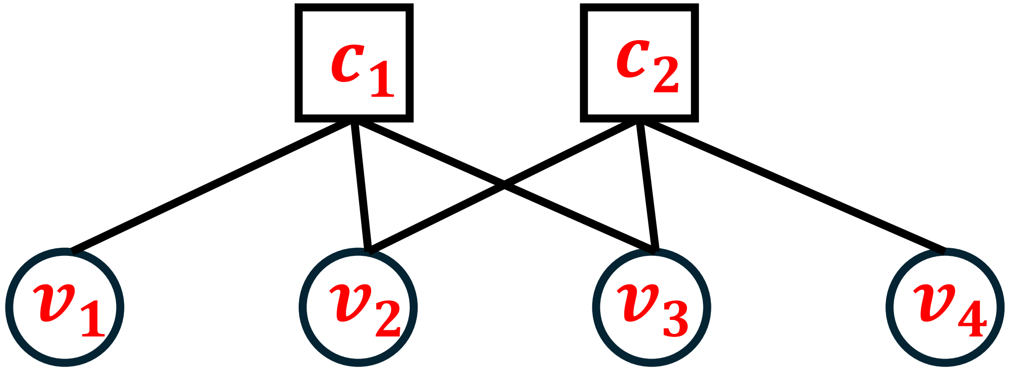
Hypergraph Product Codes

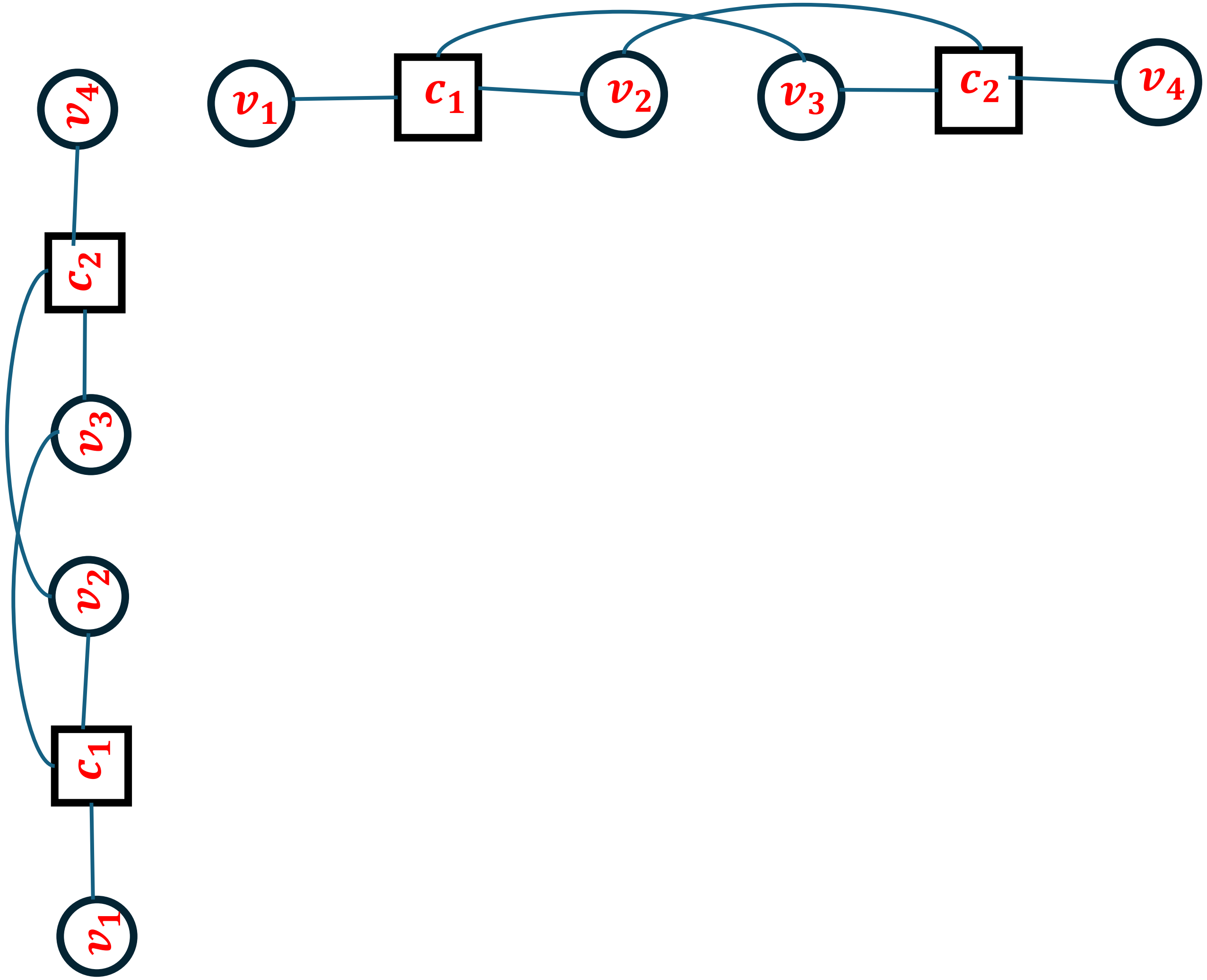
- Combine two classical binary linear codes to obtain a quantum code

Hypergraph Product Codes

- Combine two classical binary linear codes to obtain a quantum code

$$
\begin{array}{c c}
H = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} & \begin{array}{c} c_1 \\ c_2 \end{array} \\
\begin{array}{cccc} \quad v_1 & v_2 & v_3 & v_4 \end{array} &
\end{array}
$$

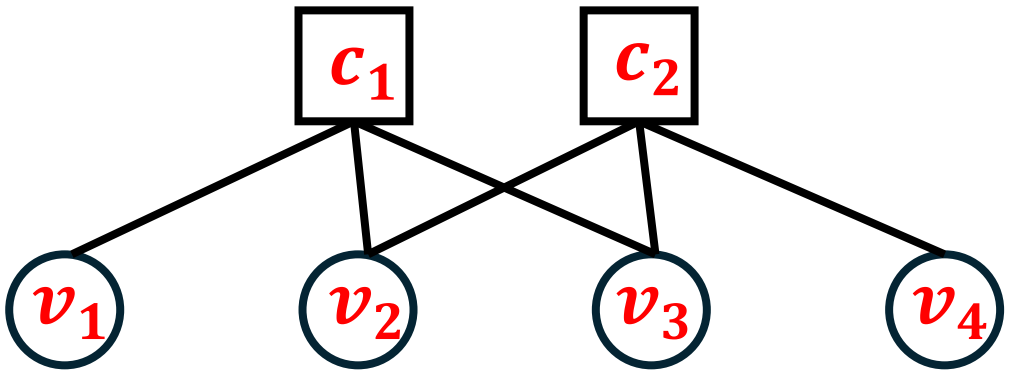


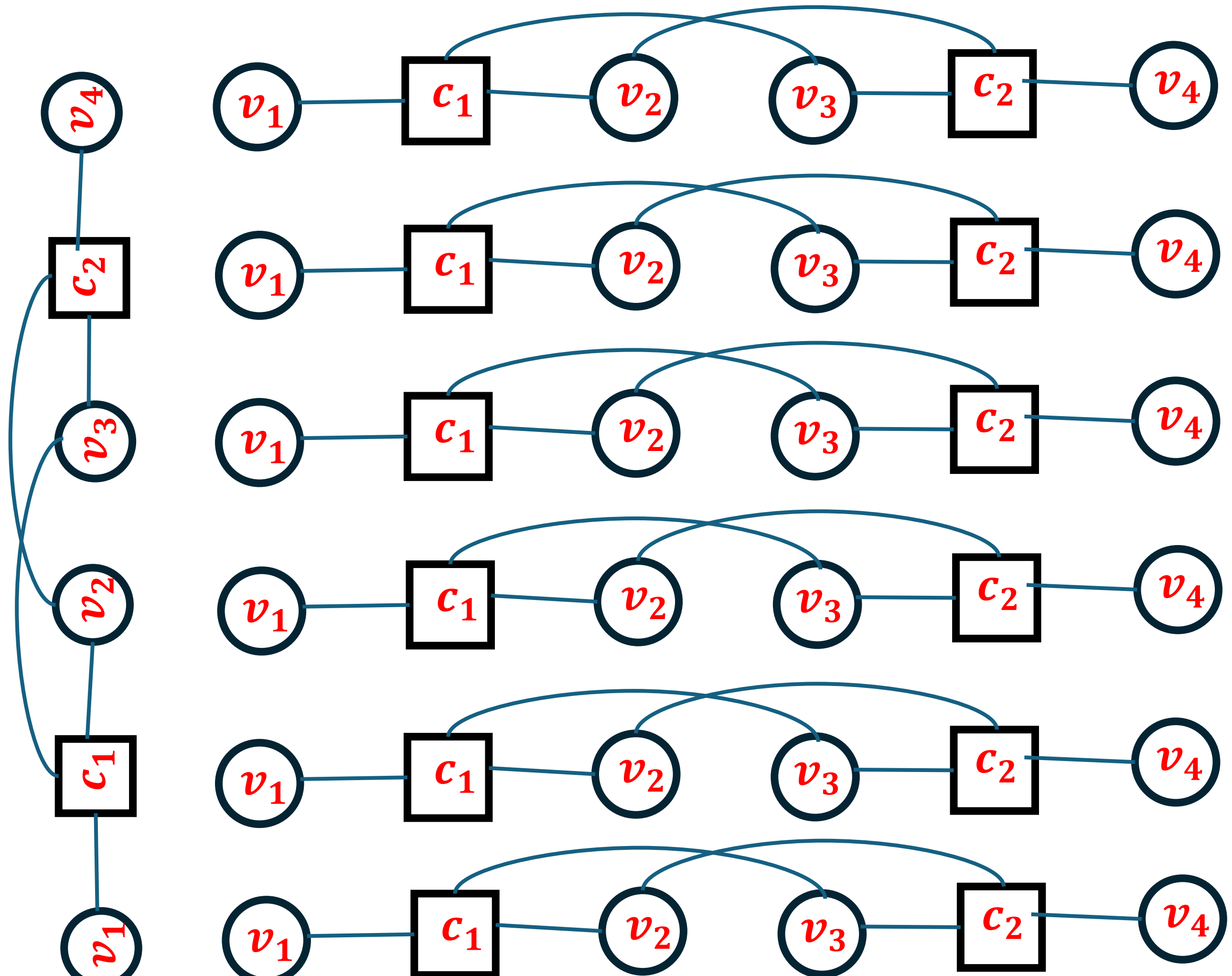


Hypergraph Product Codes

- Combine two classical binary linear codes to obtain a quantum code

$$\begin{array}{c} H = \end{array} \begin{array}{c|c} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} & \begin{array}{c} c_1 \\ c_2 \end{array} \\ \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} & \end{array}$$

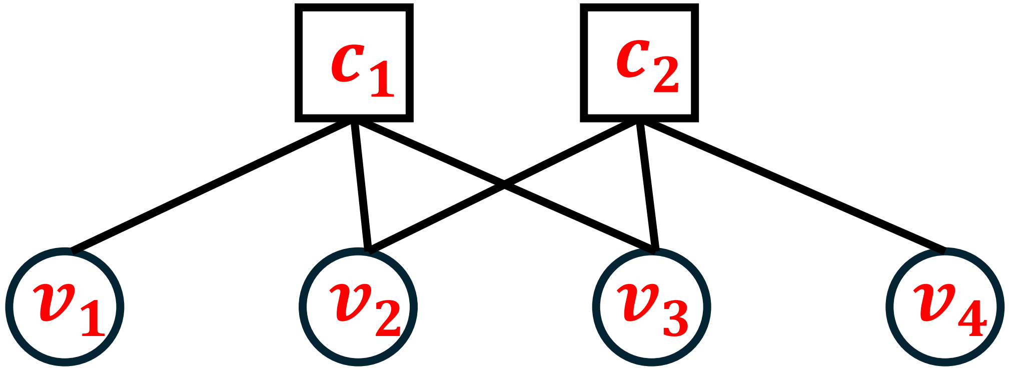


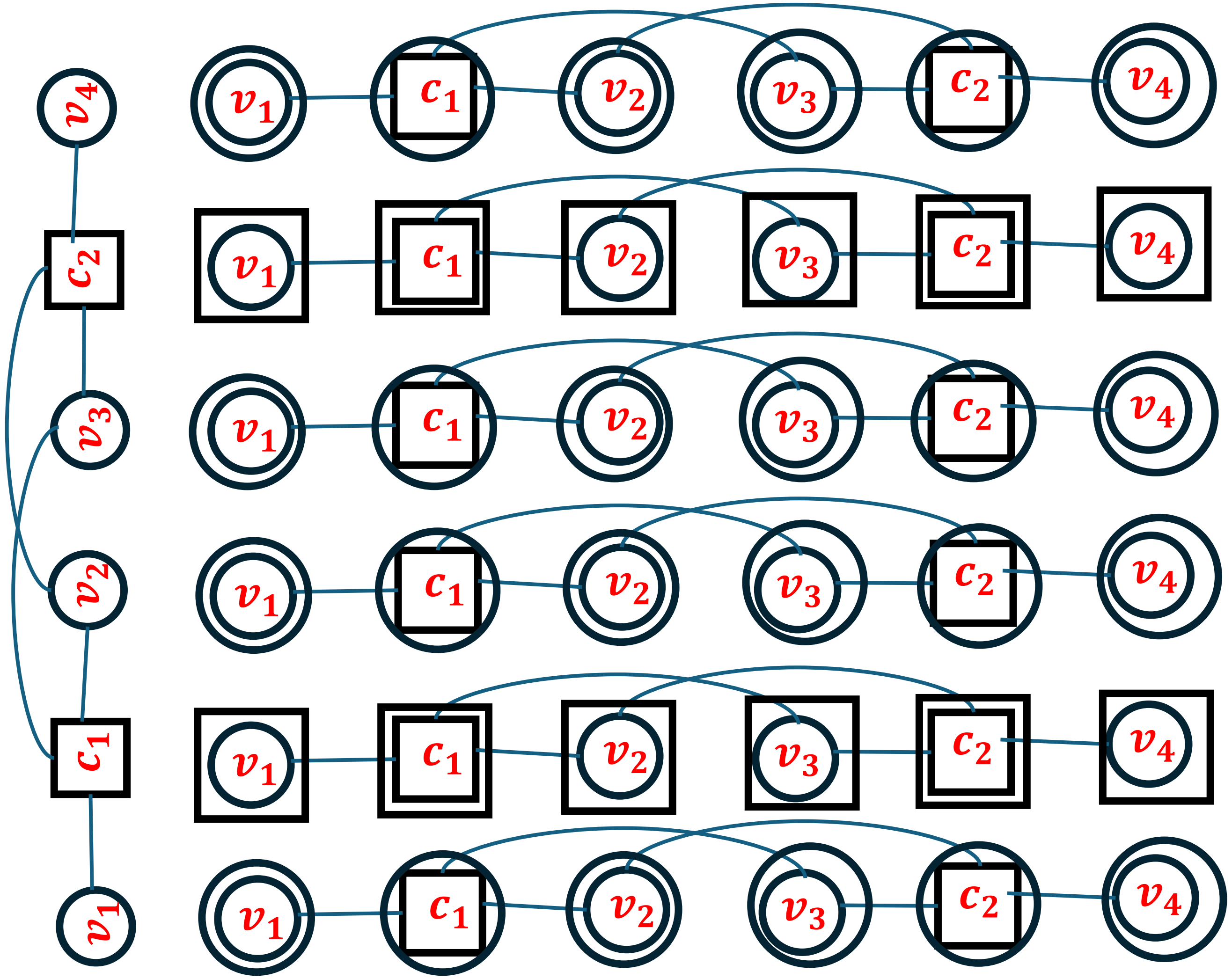


Hypergraph Product Codes

- Combine two classical binary linear codes to obtain a quantum code

$$
\begin{array}{cc}
H = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} & \begin{array}{c} c_1 \\ c_2 \end{array} \\
\begin{array}{cccc} \quad v_1 & v_2 & v_3 & v_4 \end{array} &
\end{array}
$$

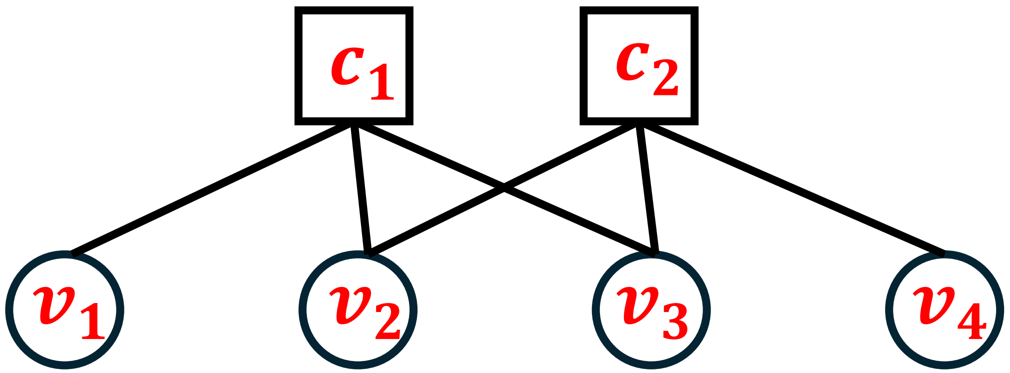


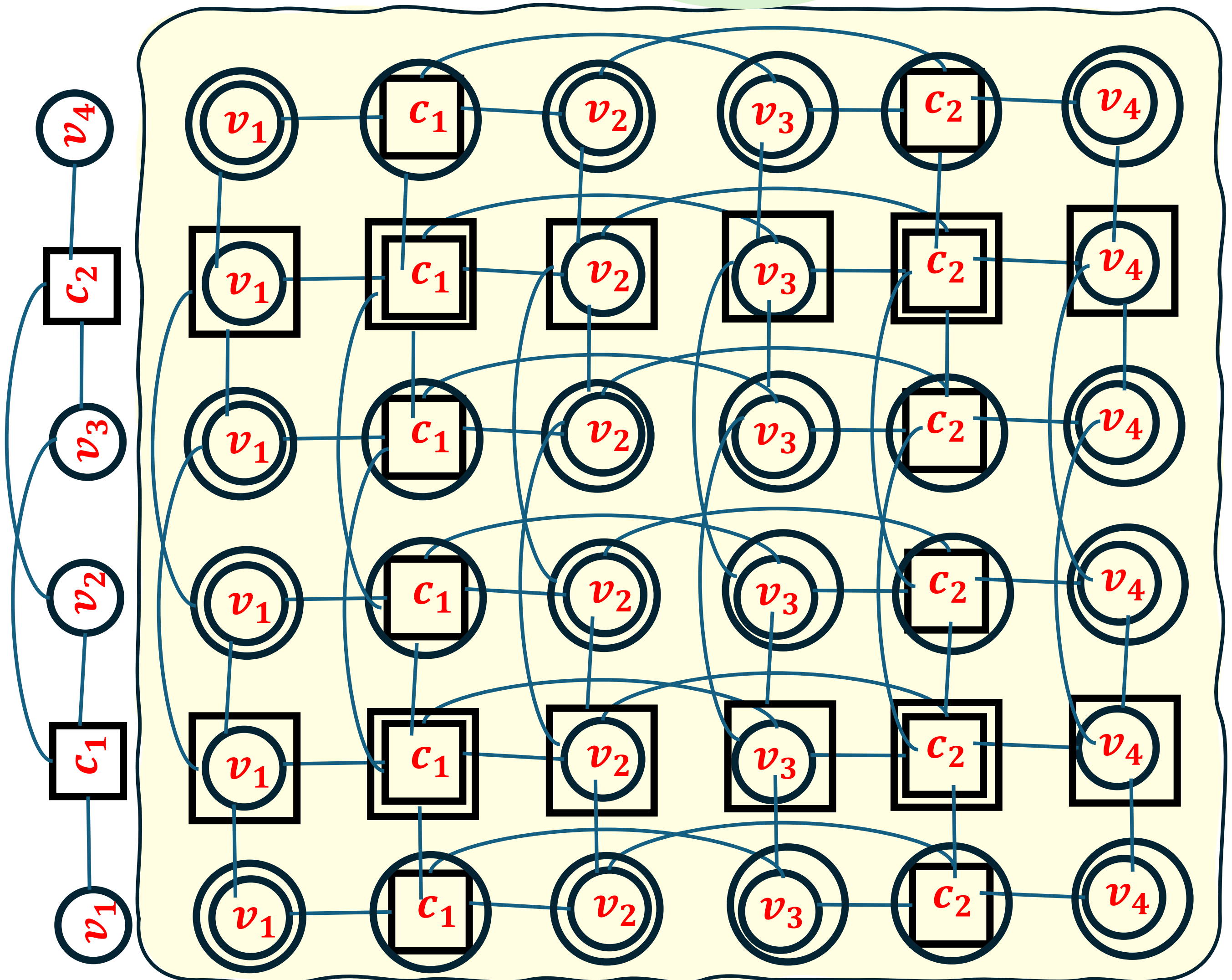


Hypergraph Product Codes

- Combine two classical binary linear codes to obtain a quantum code

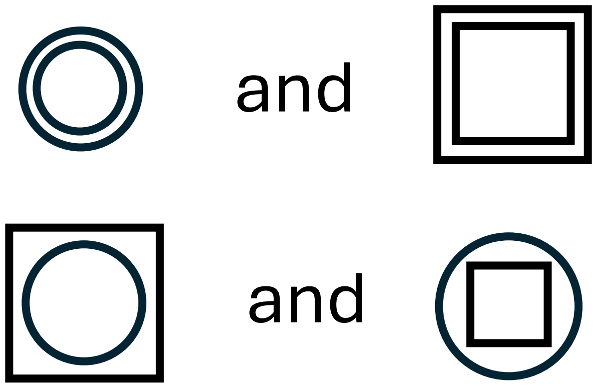
$$\begin{array}{c|c} H = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} & \begin{array}{c} c_1 \\ c_2 \end{array} \\ \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} & \end{array}$$





Hypergraph Product Codes

- Hypergraph product (quantum) code: $(n,k)\times(n,k) \mapsto [[n_1 n_2, k_1 k_2]]$



qubits

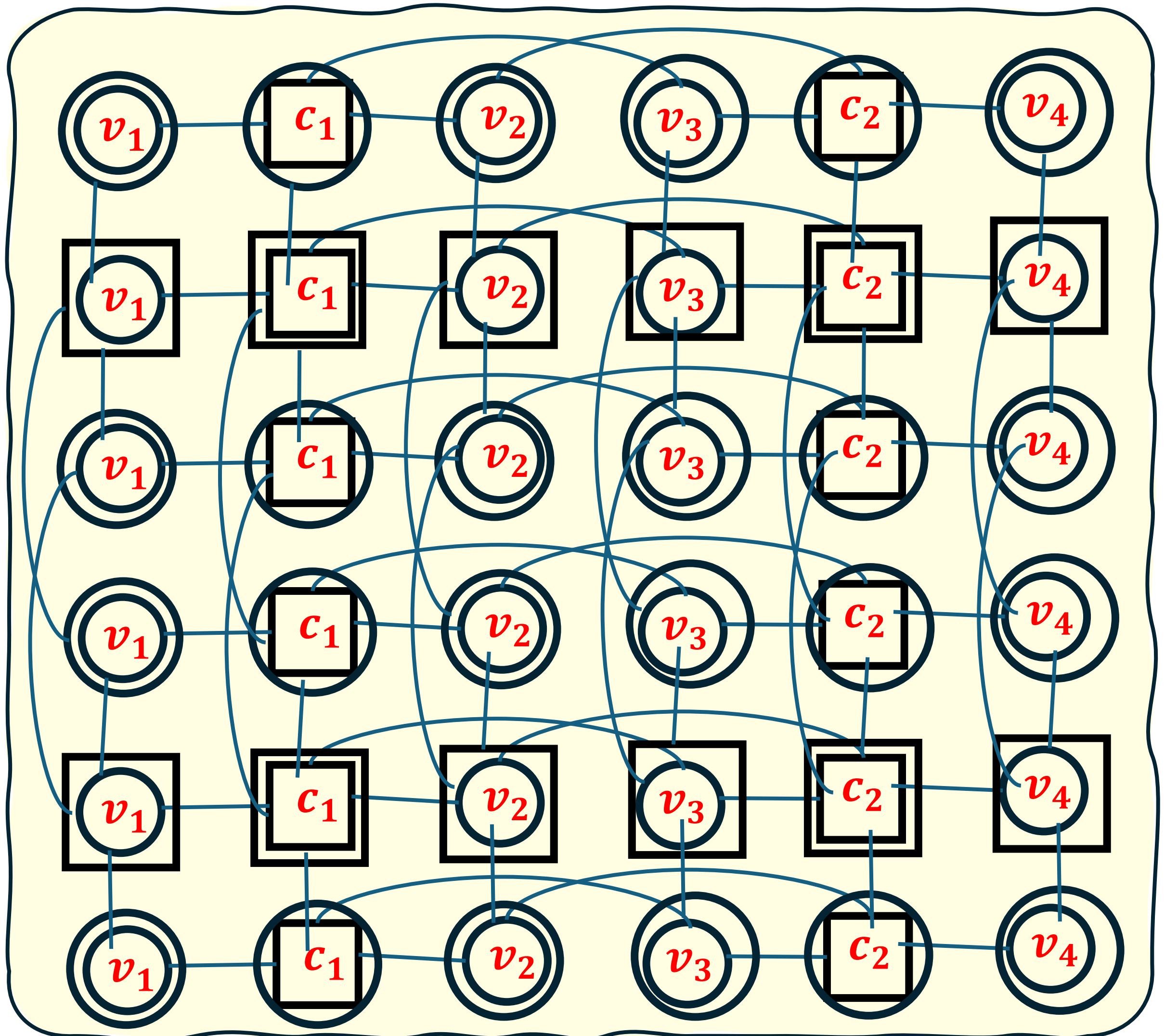
Stabilizer Generators

X-type **Z-type**

$$H_Q = \begin{bmatrix} H \otimes I_{n\times n} & I_{m\times m} \otimes H^T \\ I_{n\times n} \otimes H & H^T \otimes I_{m\times m} \end{bmatrix}$$

$H : (m \times n)$ parity check matrix

physical qubits: $n^2 + m^2$, # logical qubits: k^2



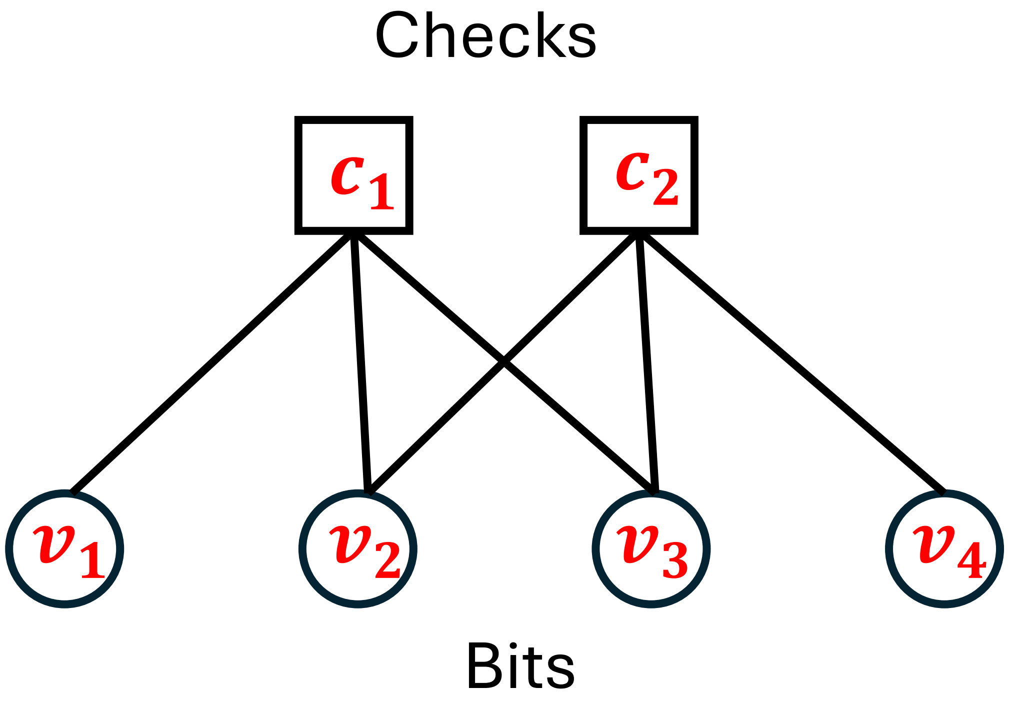
Belief Propagation (BP) Decoders

- Linear time decoder for any (q)LDPC code for a stochastic error model : (p_x, p_z).
- Classical BP algorithm independently on the Tanner Graphs describing X and Z stabilizers.

Belief Propagation (BP) Decoders

- Linear time decoder for any (q)LDPC code for a stochastic error model : (p_x, p_z).
- Classical BP algorithm independently on the Tanner Graphs describing X and Z stabilizers.

$$H = \begin{array}{c} \left[\begin{array}{cccc} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{array}\right] \\ \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \end{array} \begin{array}{c} c_1 \\ c_2 \\ \end{array}$$



Belief Propagation (BP) Decoders

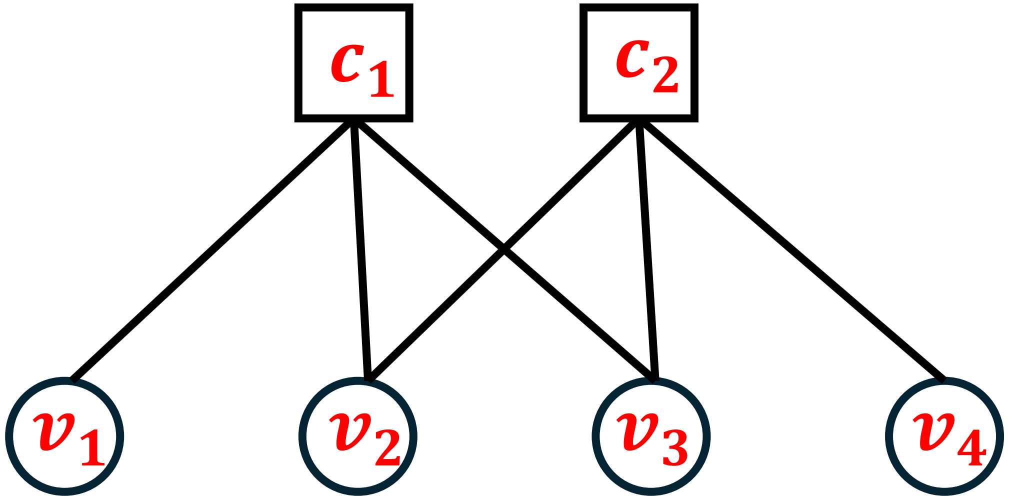
- Linear time decoder for any (q)LDPC code for a stochastic error model : (p_x, p_z).
- Classical BP algorithm independently on the Tanner Graphs describing X and Z stabilizers.

$$H = \begin{array}{cc} \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} & \begin{array}{c} c_1 \\ c_2 \end{array} \\ \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} & \end{array}$$

Log Likelihood Ratio $= l_v = \log\left(\frac{1-p_{x/z}}{p_{x/z}}\right)$

Probability of X (Z) error on bit v

Checks



Bits

Belief Propagation (BP) Decoders

- Linear time decoder for any (q)LDPC code for a stochastic error model : (p_x, p_z).
- Classical BP algorithm independently on the Tanner Graphs describing X and Z stabilizers.

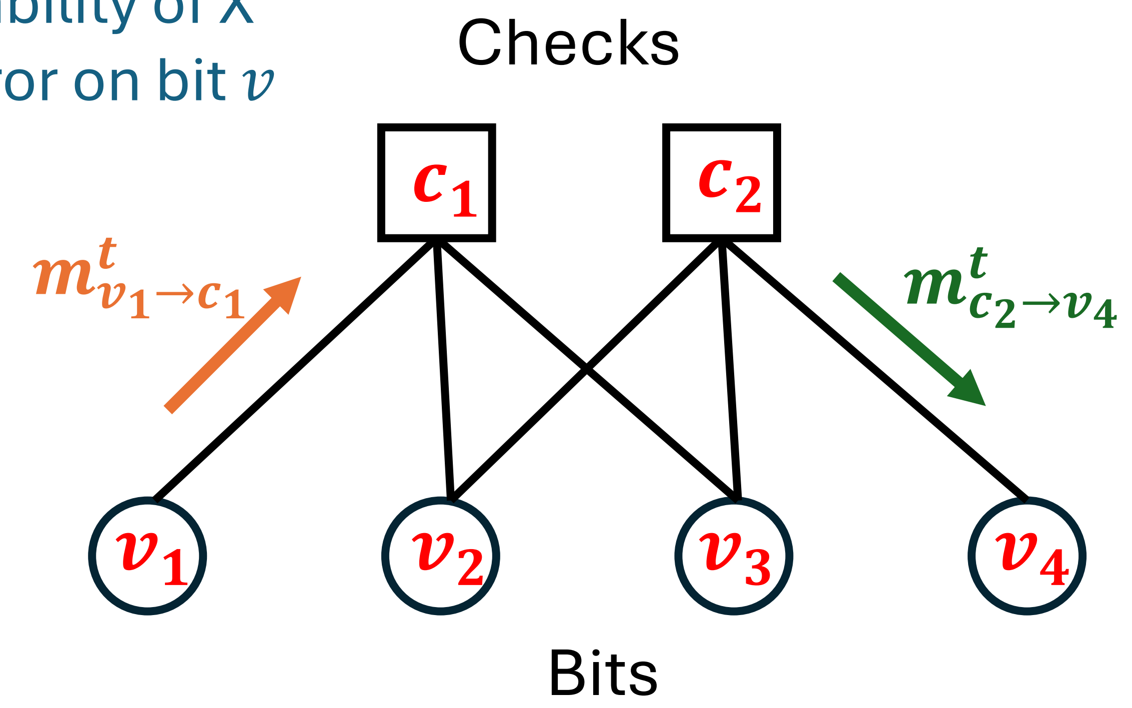
$$\begin{array}{cc} H = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} & \begin{array}{c} \boldsymbol{c_1} \\ \boldsymbol{c_2} \end{array} \\ \quad\;\; \begin{array}{cccc} \boldsymbol{v_1} & \boldsymbol{v_2} & \boldsymbol{v_3} & \boldsymbol{v_4} \end{array} & \end{array}$$

Log Likelihood Ratio $= \mathrm{l_v} = \log\left(\frac{1-p_{x/z}}{p_{x/z}}\right)$

Probability of X (Z) error on bit v

$$\boldsymbol{m_{c\to v}^{t} = 2\,(-1)^{s_c}\tanh^{-1}\left(\prod_{v'\in N(c)} \tanh\left(\frac{m_{v'\to c}^{t}}{2}\right)\right)}$$

$$\boldsymbol{m_{v\to c}^{t+1} = l_v + \sum_{c'\in N(v)\backslash c} m_{v'\to c}^{t}}$$



BP as a Neural Network

Log Likelihood Ratio $= \mathrm{l_v} = \log\left(\frac{1-p_{x/z}}{p_{x/z}}\right)$

Repeat for T iterations

$$\log\tanh\left(\frac{m^t_{c\to v}}{2}\right) = i\,\pi\,s_c + \sum_{v'\in N(c)\setminus \mathrm{v}} \log\tanh\left(\frac{m^t_{v'\to c}}{2}\right)$$

$$m^{t+1}_{v\to c} = l_v + \sum_{c'\in N(v)\setminus \mathrm{c}} m^t_{v'\to c}$$

Compute refined beliefs

$$\mu^T_{\mathrm{v}} = l_v + \sum_{c\in N(v)} m^T_{c\to v}$$

$\mu^T_v > 0$: *no error*

$\mu^T_v < 0$: *error*

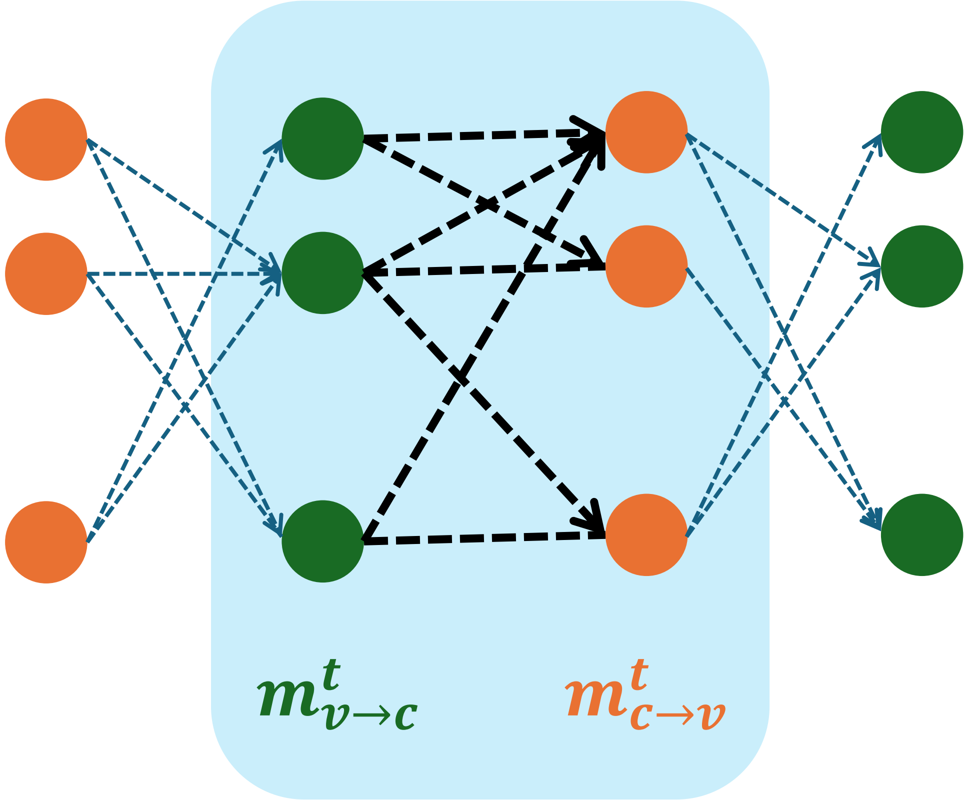
BP as a Neural Network

Log Likelihood Ratio $= \mathrm{l_v} = \log\left(\frac{1-p_{x/z}}{p_{x/z}}\right)$

Repeat for T iterations

$$\log\tanh\left(\frac{m^t_{c\to v}}{2}\right) = i\,\pi\,s_c + \sum_{v'\in N(c)\backslash \mathrm{v}} \log\tanh\left(\frac{m^t_{v'\to c}}{2}\right)$$

$$m^{t+1}_{v\to c} = l_v + \sum_{c'\in N(v)\backslash \mathrm{c}} m^t_{v'\to c}$$

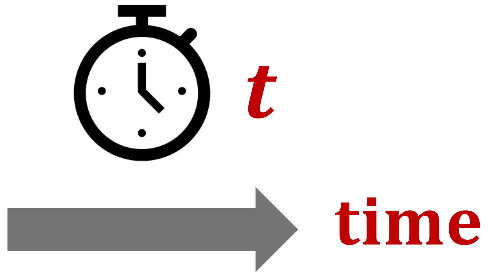


Compute refined beliefs

$$\mu^T_{\mathrm{v}} = l_v + \sum_{c\in N(v)} m^T_{c\to v}$$

$\mu^T_v > 0$: *no error*

$\mu^T_v < 0$: *error*



BP as a Neural Network

Log Likelihood Ratio $= l_v = \log\left(\frac{1-p_{x/z}}{p_{x/z}}\right)$

Repeat for T iterations

$$\log\tanh\left(\frac{m_{c\to v}^{t}}{2}\right) = i\,\pi\,s_c + \sum_{v'\in N(c)\setminus v} \log\tanh\left(\frac{m_{v'\to c}^{t}}{2}\right)$$

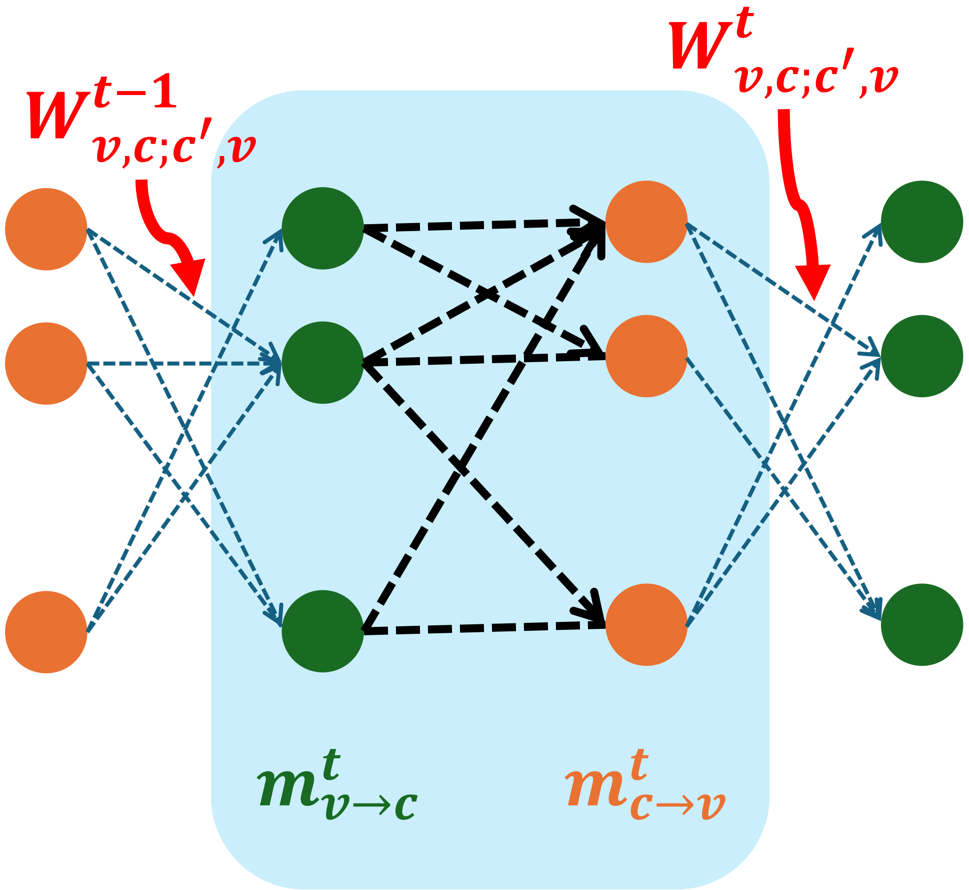
$$m_{v\to c}^{t+1} = b_v\, l_v + \sum_{c'\in N(v)\setminus c} W_{v,c;c',v}^{t}\, m_{v'\to c}^{t}$$

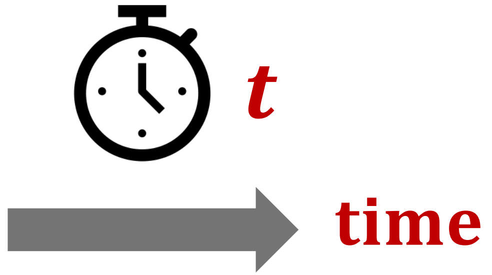
Compute refined beliefs

$$\mu_v^T = b_v l_v + \sum_{c\in N(v)} W_{v;c',v}^{T}\, m_{c\to v}^{T}$$

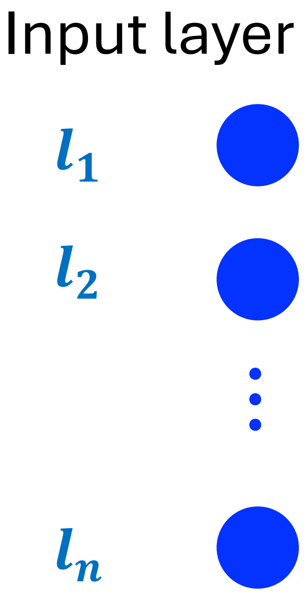
$\mu_v^T > 0$: no error

$\mu_v^T < 0$: error





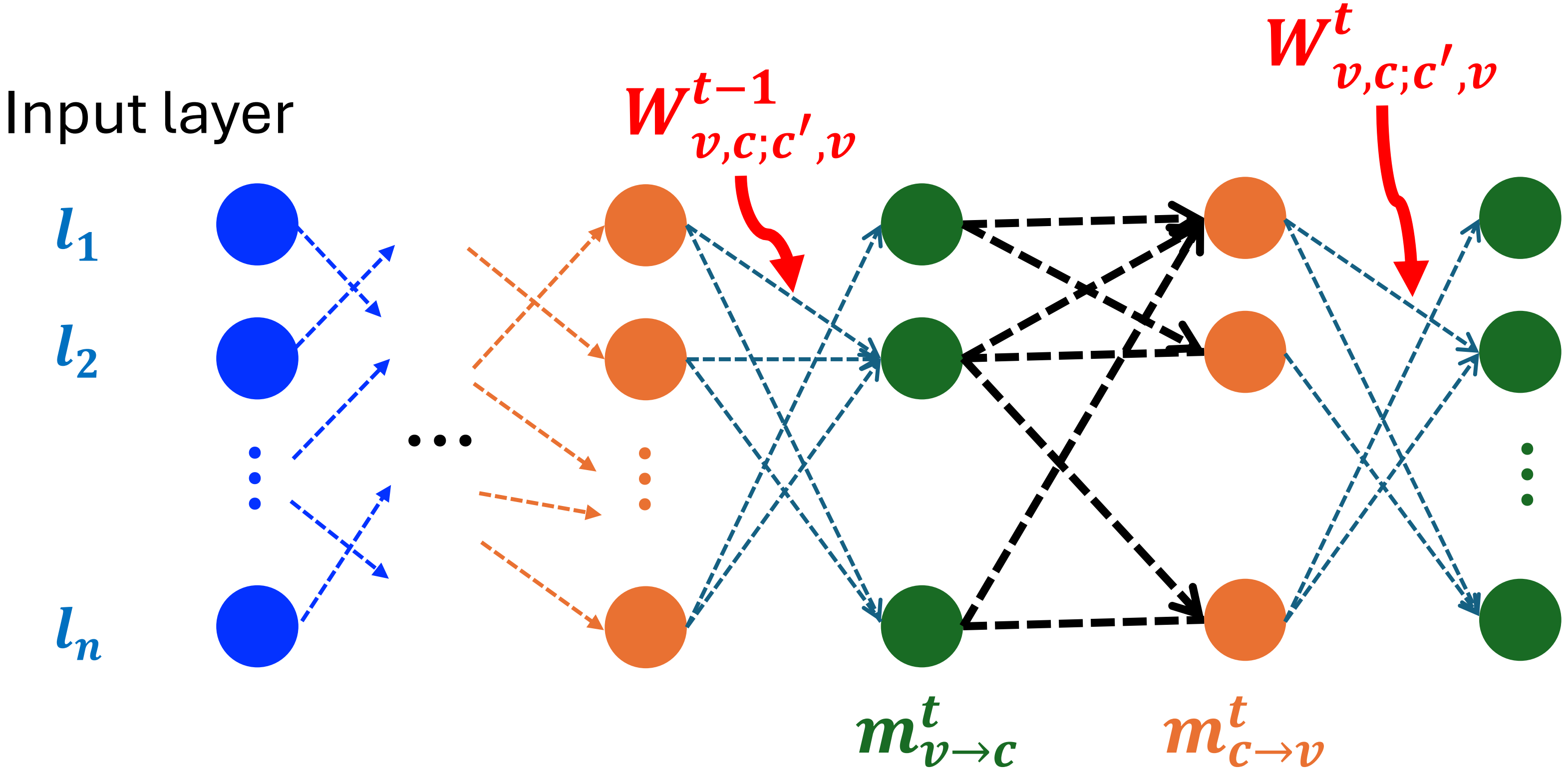
BP as a Neural Network



Nachmani et. al., 2016. IEEE (Allerton)

Liu, Poulin, 2019. Phys. Rev. Lett. 122, 200501

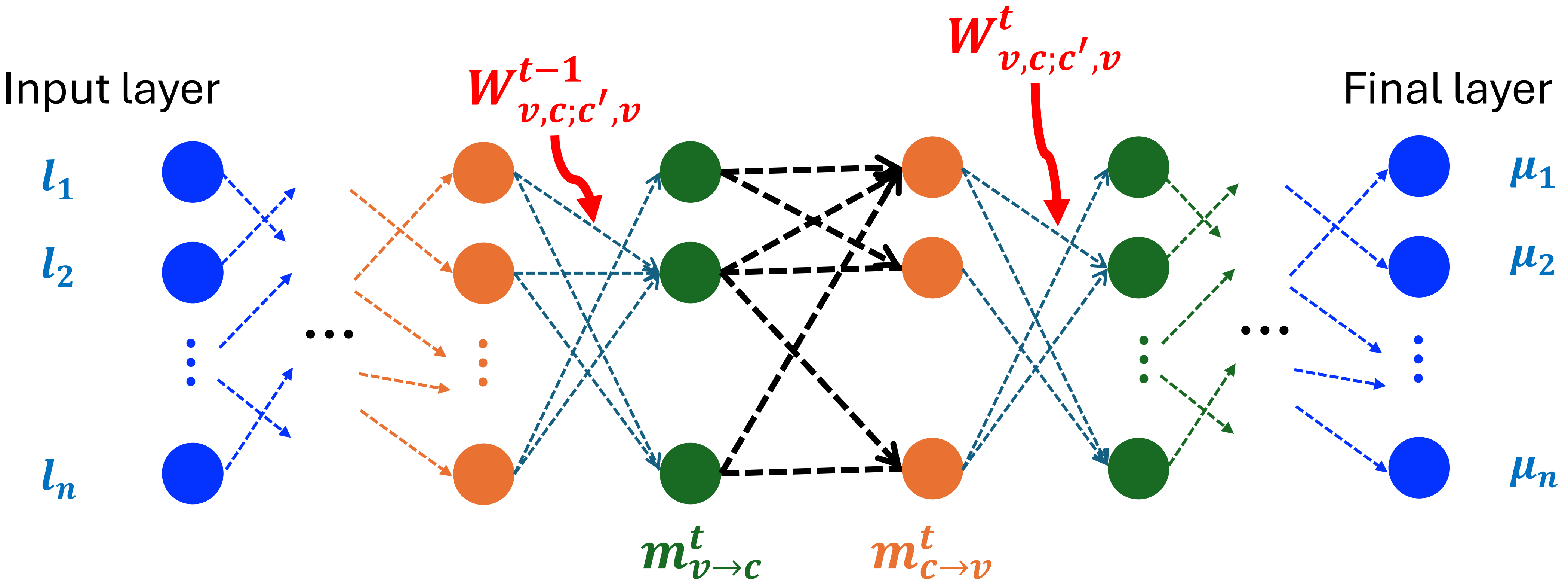
BP as a Neural Network



Nachmani et. al., 2016. IEEE (Allerton)

Liu, Poulin, 2019. Phys. Rev. Lett. 122, 200501

BP as a Neural Network



Activation function at layers

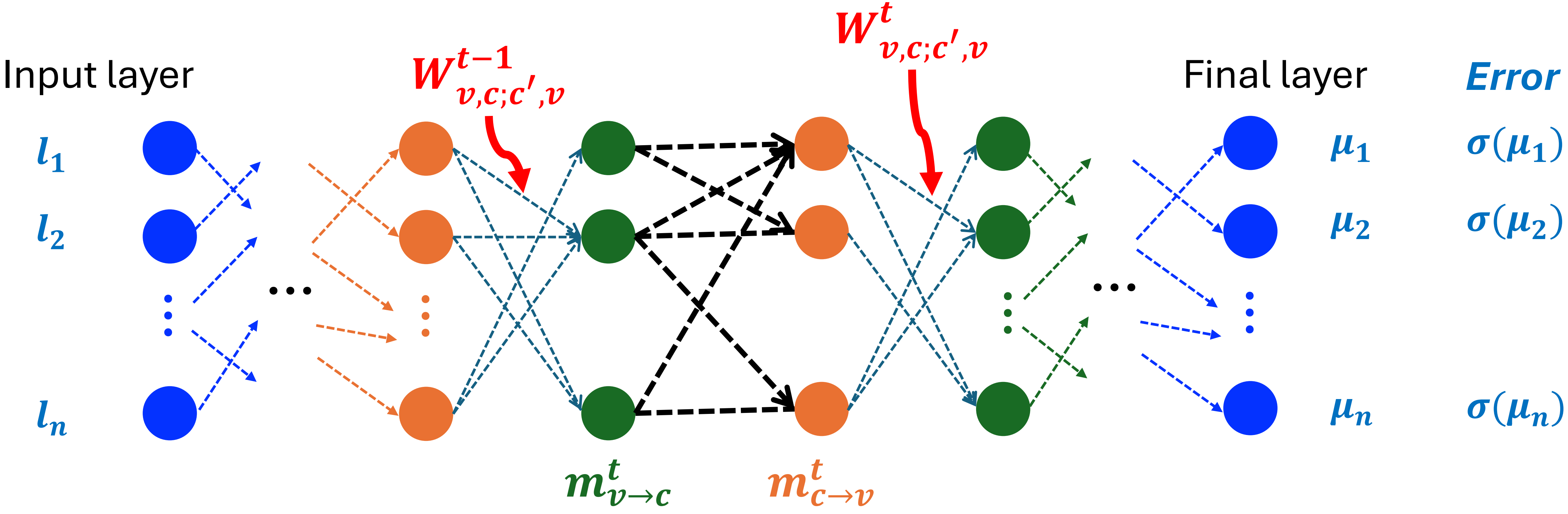
$$a(x) = 2\tanh^{-1}(e^x)$$

$$a(x) = \log\tanh(x/2)$$

Nachmani et. al., 2016. IEEE (Allerton)

Liu, Poulin, 2019. Phys. Rev. Lett. 122, 200501

BP as a Neural Network



Activation function at layers

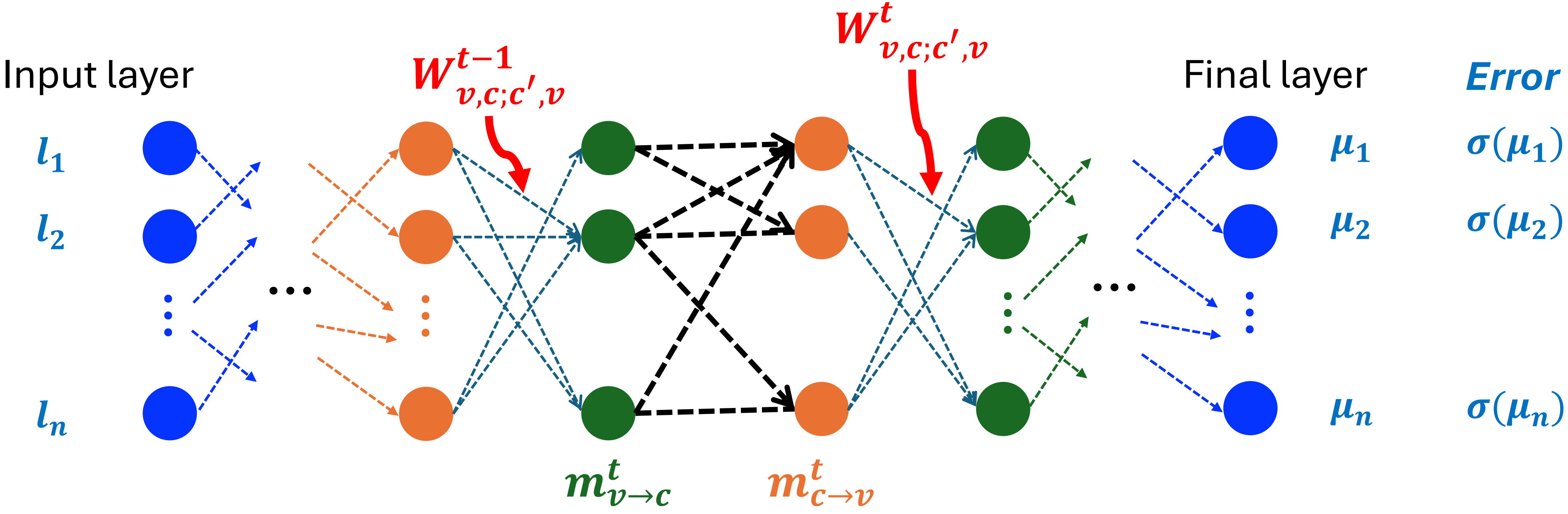
$$a(x) = 2\tanh^{-1}(e^x)$$

$$a(x) = \log\tanh(x/2)$$

Nachmani et. al., 2016. IEEE (Allerton)

Liu, Poulin, 2019. Phys. Rev. Lett. 122, 200501

BP as a Neural Network



Activation function at layers

$a(x) = 2\tanh^{-1}(e^x)$

$a(x) = \log\tanh(x/2)$

Weights obtaining from training the NN:

$s = H^{\perp}(\sigma(\mu_v) + e)$ $Loss = |\sin(\pi\, s_c/2)|.$

Nachmani et. al., 2016. IEEE (Allerton)

Liu, Poulin, 2019. Phys. Rev. Lett. 122, 200501

Adding CER inputs to BP

$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \, \rho \, P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

Adding CER inputs to BP

$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \ \rho \ P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

Two-qubit correlations: spatial cross-talk, frequency coupling (imperfect two-qubit gates)

Single qubit error rates

$p_{i,I}, p_{i,X}, p_{i,Z}, p_{i,Y}$: on each qubit

Two-qubit error rates

$p_{ij,II}, p_{i,XX}, \ldots, p_{ZZ}$: on qubit pairs

Fazio et al 2026. Phys. Rev. X 16, 011030

Harper and Flammia, 2023 . PRXQuantum.4.040311

Adding CER inputs to BP

Memory noise:

$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \ \rho \ P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

Two-qubit correlations: spatial cross-talk, frequency coupling (imperfect two-qubit gates)

Single qubit error rates

$p_{i,I}, p_{i,X}, p_{i,Z}, p_{i,Y}$: on each qubit

Two-qubit error rates

$p_{ij,II}, p_{i,XX}, \ldots, p_{ZZ}$: on qubit pairs

Weights in Neural BP: $W^t_{v,c;c',v}$ and b_v : minimize Loss per sample

$$\sum_c f\left(H_\perp(e + \sigma(\mu))_c\right)$$

Fazio et al 2026. Phys. Rev. X 16, 011030

Harper and Flammia, 2023 . PRXQuantum.4.040311

Adding CER inputs to BP

Memory noise:

$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \ \rho \ P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

Two-qubit correlations: spatial cross-talk, frequency coupling (imperfect two-qubit gates)

Single qubit error rates

$p_{i,I}, p_{i,X}, p_{i,Z}, p_{i,Y}$: on each qubit

Two-qubit error rates

$p_{ij,II}, p_{i,XX}, \dots, p_{ZZ}$: on qubit pairs

Weights in Neural BP: $W^t_{v,c;c',v}$ and b_v : minimize Loss per sample

$$\sum_c f\left(H_\perp(e + \sigma(\mu))_c\right) + \sum_{\langle i,j\rangle \in C} \lambda_{i,j} \ \sigma(\mu_i^T) \left(1 - \sigma(\mu_i^T)\right)$$

Penalize configurations where error occurs on qubit i and no error on j.

Connectivity graph (hardware)

Fazio et al 2026. Phys. Rev. X 16, 011030

Harper and Flammia, 2023 . PRXQuantum.4.040311

Adding CER inputs to BP

Memory noise:

$$\mathcal{E}(\rho) = \sum_{i,j} \chi_{i,j} P_i \ \rho \ P_j$$

Pauli decay rates: $\chi_{i,i} \geq 0$

Two-qubit correlations: spatial cross-talk, frequency coupling (imperfect two-qubit gates)

Single qubit error rates

$p_{i,I}, p_{i,X}, p_{i,Z}, p_{i,Y}$: on each qubit

Two-qubit error rates

$p_{ij,II}, p_{i,XX}, \ldots, p_{ZZ}$: on qubit pairs

Weights in Neural BP: $W^t_{v,c;c',v}$ and b_v : minimize Loss per sample

$$\sum_c f\left(H_\perp(e + \sigma(\mu))_c\right) + \sum_{\langle i,j\rangle \in C} \lambda_{i,j} \ \sigma(\mu_i^T) \left(1 - \sigma(\mu_i^T)\right)$$

Penalize configurations where error occurs on qubit i and no error on j.

Connectivity graph (hardware)

From CER

Fazio et al 2026. Phys. Rev. X 16, 011030

Harper and Flammia, 2023 . PRXQuantum.4.040311

Numerical Results

Neural BP decoder vs. *BP-OSD decoder*

for correlated error models. (Monte Carlo)

Nickerson and Brown, 2019, Quantum

Roffe et. al. Phys. Rev. Research 2, 043423, 2020

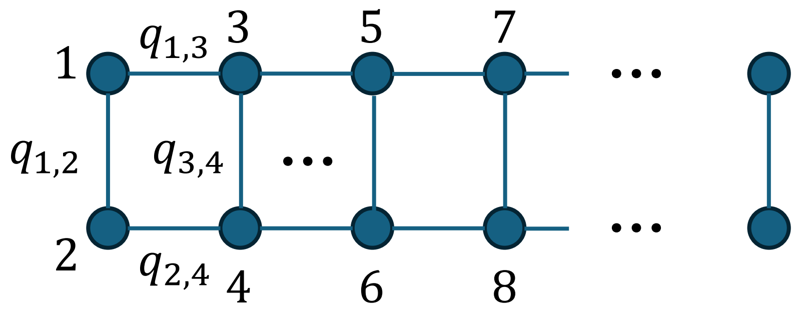
Numerical Results

Neural BP decoder vs. *BP-OSD decoder*

for correlated error models. (Monte Carlo)

Error model: Ballistic Error Model

- Each qubit independently undergoes an error (X/Z) with probability p.
- Qubit i has an error ⇒ each connected qubit j has error with probability $q_{i,j}$.



- CER provides two qubit marginals

Nickerson and Brown, 2019, Quantum

Roffe et. al. Phys. Rev. Research 2, 043423, 2020

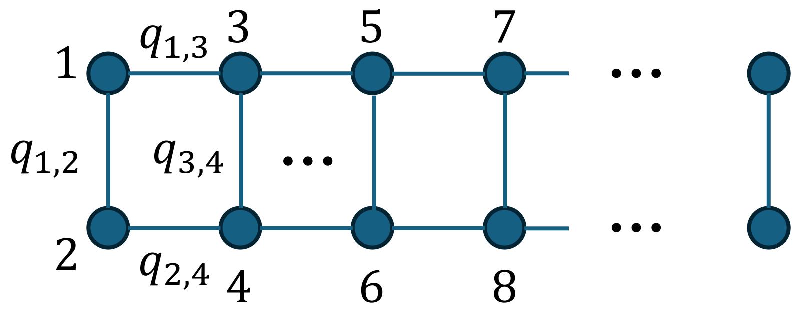
Numerical Results

Neural BP decoder vs. *BP-OSD decoder*

for correlated error models. (Monte Carlo)

***Error model:* Ballistic Error Model**

- Each qubit independently undergoes an error (X/Z) with probability p.
- Qubit i has an error $\Rightarrow$ each connected qubit j has error with probability $q_{i,j}$.



- CER provides two qubit marginals

***Code:* HGP of Hamming Codes: [[58,16,9]]**

Nickerson and Brown, 2019, Quantum

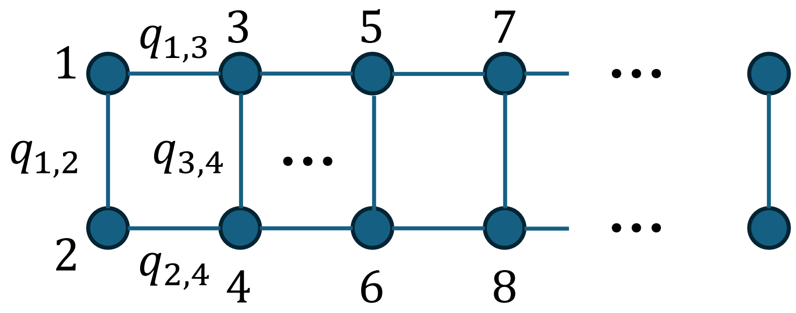
Roffe et. al. Phys. Rev. Research 2, 043423, 2020

Numerical Results

Neural BP decoder vs. *BP-OSD decoder* for correlated error models. (Monte Carlo)

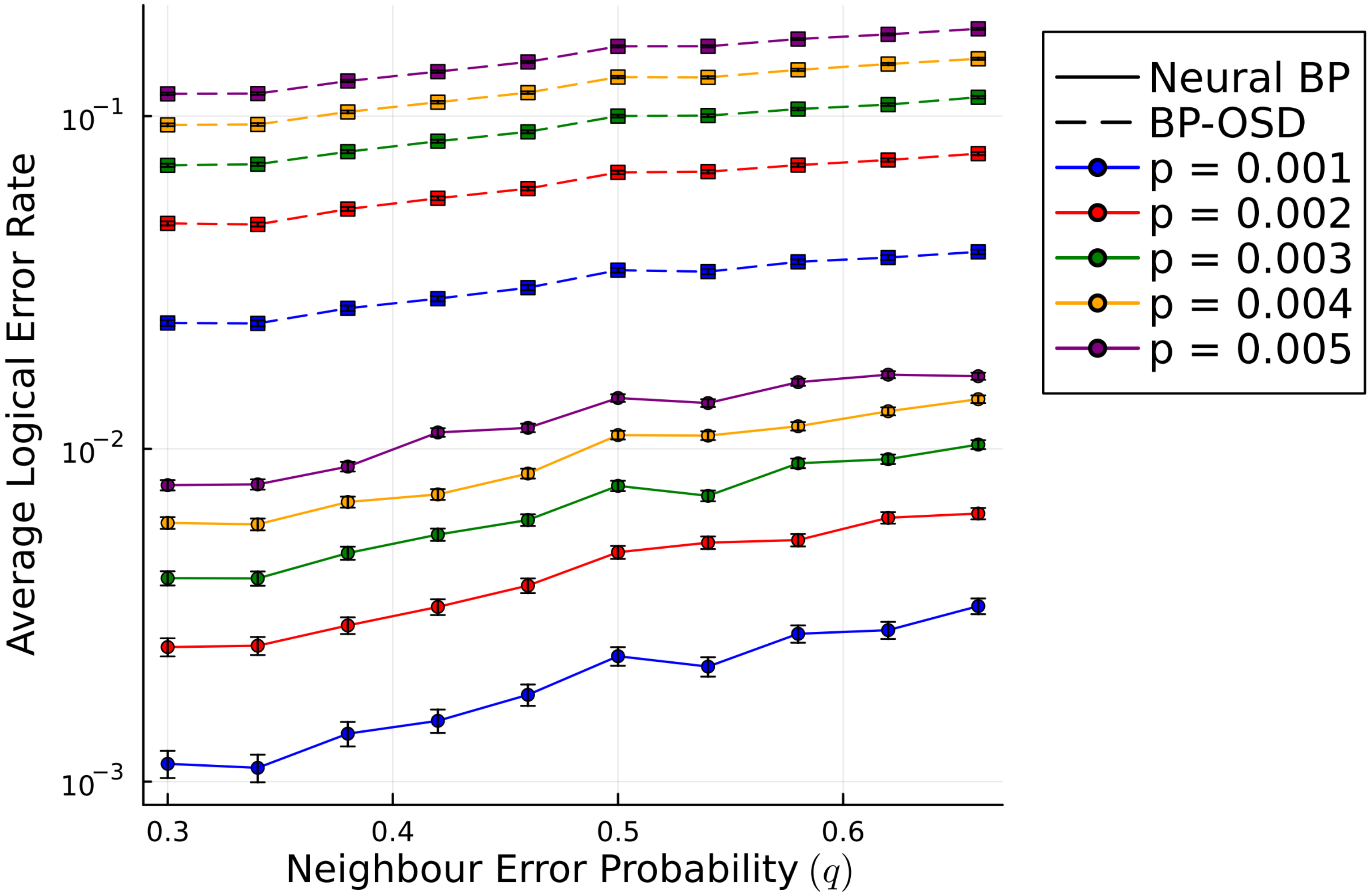
Error model: Ballistic Error Model

- Each qubit independently undergoes an error (X/Z) with probability p.
- Qubit i has an error $\Rightarrow$ each connected qubit j has error with probability $q_{i,j}$.



- CER provides two qubit marginals

Code: HGP of Hamming Codes: [[58,16,9]]



Nickerson and Brown, 2019, Quantum

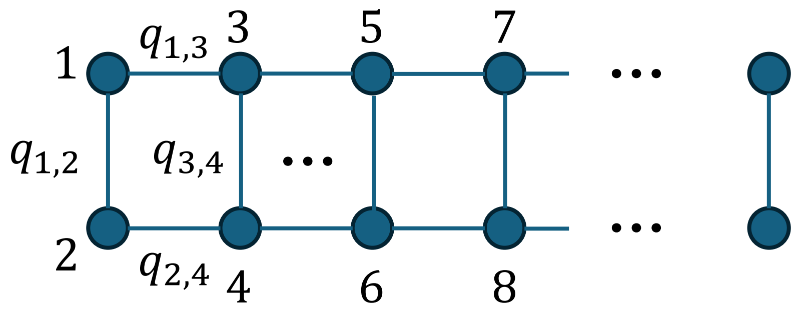
Roffe et. al. Phys. Rev. Research 2, 043423, 2020

Numerical Results

Neural BP decoder vs. *BP-OSD decoder* for correlated error models. (Monte Carlo)

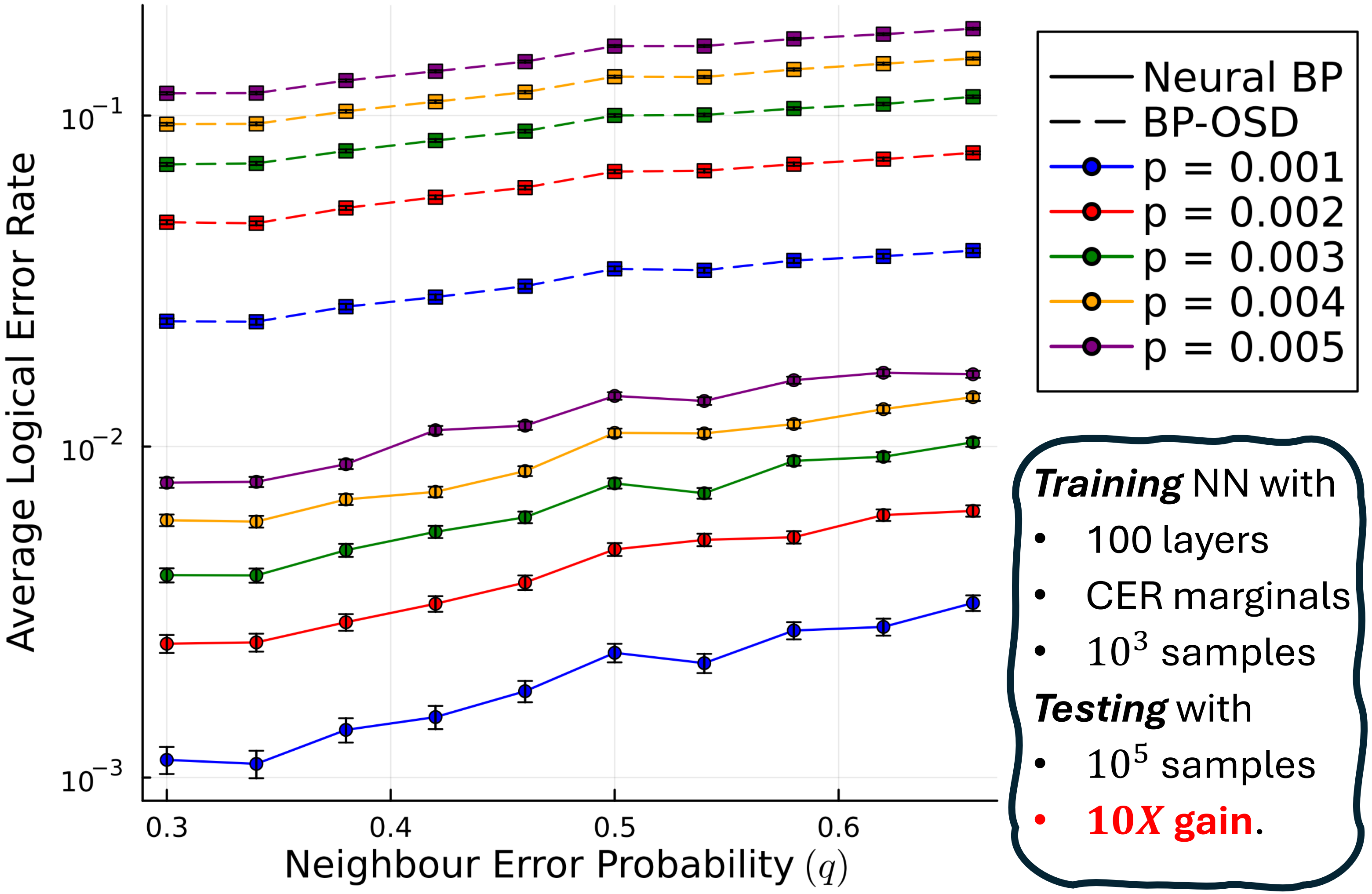
Error model: ***Ballistic Error Model***

- Each qubit independently undergoes an error (X/Z) with probability p.
- Qubit i has an error $\Rightarrow$ each connected qubit j has error with probability $q_{i,j}$.



- CER provides two qubit marginals

Code: ***HGP of Hamming Codes: [[58,16,9]]***



Training NN with
- 100 layers
- CER marginals
- 10^3 samples

Testing with
- 10^5 samples
- **$10X$ gain.**

Nickerson and Brown, 2019, Quantum

Roffe et. al. Phys. Rev. Research 2, 043423, 2020

Conclusions and future work

- What we have demonstrated:
  - Decoding quantum LDPC codes against correlated errors.
  - Unravel BP into a Neural Network and optimize weights to adapt to correlations
  - $10\mathrm{X}$ gain for HGP under Ballistic Noise, an instance of correlated error model.

Conclusions and future work

- What we have demonstrated:
  - Decoding quantum LDPC codes against correlated errors.
  - Unravel BP into a Neural Network and optimize weights to adapt to correlations
  - $10X$ gain for HGP under Ballistic Noise, an instance of correlated error model.

- Future work and similar strategies:
  - Improving threshold for other family of quantum LDPC codes (La-Cross, Bicycle).
  - Correlated errors from Hamiltonian dynamics: beyond the Pauli paradigm.
  - Soft constraints and BP with higher dimensional messages.